

A Project Proposal for Online Code Editor





Introduction

Websocket is an online code editor project is a cutting-edge web-based development tool designed to provide a seamless and collaborative coding experience for developers, students, and professionals.

This platform aims to simplify coding tasks by offering a feature-rich, accessible, and user-friendly environment where users can write, compile, debug, and share code effortlessly.



The Problem with Traditional Online Code Editors

Traditional online code editors lack real-time collaboration, which slows down the development process. Furthermore, most online code editors are limited in terms of functionality and customization, making it difficult for developers to work efficiently.



KEY FEATURES :

Code Editing: users can write, edit, and format code in various programming

Multi-Language Support: the editor supports a wide range of programming languages.

Live Preview: instantly visualise the output of your code with a live preview feature,

Collaboration Tools: real-time collaboration features allow multiple users to code together.

Auto-Save: automatically saving work to prevent data loss.





Technology Stack :

FRONTEND:

HTML5

CSS3

JavaScript (React Js.)

BACKEND:

Node Js

Express Js

DATABASE:

MongoDB

REAL-TIME FEATURE:

WebSocket

Implementation Plan

To implement a WebSocket-based online code editor, we will need to develop a custom editor that supports real-time collaboration. We will also need to integrate WebSocket technology into the editor and create a server that can handle multiple connections.



FUTURE ENHANCEMENTS:

- AI Code Assistance: Implement ai-driven code suggestions and auto-completion.
- Mobile App: Develop a mobile version for coding on the go.
- Integrated Documentation: Provide instant access to documentation for various languages and libraries



Conclusion

This online code editor has the potential to revolutionize online collaboration by allowing developers to work together in real-time. By increasing productivity, improving code quality, and enhancing collaboration, a WebSocket-based online code editor can help development teams achieve their goals more efficiently.

Thanks!

**Kirtan Nahar
Nimit Garg
Akshay Mann
Hardik Jain**

